

How are we to live?

Ages 16+

*Evidence-based pointers on living well —
each with its study design, its effect size, and its honest
weakness.*

Trust but verify

The method, stated plainly. Every claim on this site is presented with three things attached: its **study design**, its **effect size**, and its **honest weakness**. Every figure is tracked in a public **provenance** ledger, checked against its source and linked so you can go and read the original for yourself. This is a reference, not a campaign: no calls to action, no donate button. If it ever reads as a nice-looking set of assertions, it has failed — a confident claim with no working shown is no better than the marketing it hopes to improve on.

Trust but verify (cont.)

"Trust but verify" also sets the register for how these topics are pitched at each age — reassurance where the real risk is low, straight talk where it isn't, and never a scary number without the honest context around it.

And here is the honest boundary of the whole enterprise, put where you can see it:

Trust but verify (cont.)

This entire site assumes you **care what happens to a future version of you** — the you of ten, forty, seventy years from now. Almost every argument here is a trade: a small cost now for a larger benefit to that future person. If you genuinely do not care what happens to future-you, and you are *certain* your future self will never feel differently, then there is nothing for you here — the trade has no value if you place no value on the other side of it. We think that certainty is very hard to earn honestly, because people's future selves reliably turn out to care. But we are not going to pretend the assumption isn't there. It is the one thing this site cannot prove to you.

Smoking kills, stopping helps

Now you are old enough to weigh the long clock honestly, so here it is without softening.

Smoking kills, stopping helps (cont.)

Cigarettes are the most clearly proven cause of early death in all of medicine. Lifelong smokers die on average about **ten years younger** than people who never smoked, and roughly two in every three lifelong smokers are eventually killed by it. This is not a weak statistical hint; it is one of the most reproduced findings in science, shown in different countries, sexes and decades, with the biological mechanism traced all the way down to the exact DNA damage. The details and sources are in the evidence panel below — check them.

But the second half is what turns a grim fact into a decision you control:

Smoking kills, stopping helps (cont.)

- **The risk is not banked. It is largely reversible if you get out early.** Stopping before 40 avoids the large majority of the excess risk; stopping before 30 avoids almost all of it. Even stopping at 60 buys back years. There is no age at which quitting fails to help.
- **Initiation is the only part of the pathway fully in your hands.** Almost all lifelong smokers start before 20. Not starting removes the entire downstream problem at no cost — which is why, if you have made it this far without smoking, you have already made the highest-leverage move.
- **And if you already smoke, stopping is not a coin-flip of willpower.** Comprehensive support — behavioural help plus medication — and effective medicines (nicotine replacement, varenicline, cytisine) multiply the odds of success; the effect sizes are in the evidence below.

Smoking kills, stopping helps (cont.)

On vaping, the honest position is the same one as the younger band, just with the evidence attached: nicotine vaping delivers far fewer combustion toxicants than smoking and there is high-quality evidence it helps *established smokers* quit — for a smoker, switching is a genuine gain. For a never-smoker none of that applies: it is dependence with no offsetting benefit and no long-term safety record. (The 2019–20 lung-injury scare, EVALI, was vitamin E acetate in illicit THC cartridges, not nicotine vapes — getting that wrong is the fastest way to lose an argument you should win.)

Alcohol: no safe dose

Here is the claim in full, with its honest limits. "No safe dose" is true, but it is a claim about **the shape of the curve, not the size of the risk** — and both halves matter, because the scary half on its own would overstate how large that risk is.

Alcohol: no safe dose (cont.)

- The curve **rises from the very first drink**: there is no threshold below which risk is zero. The old "J-curve" that seemed to show light drinking protecting your heart was an artefact — the "non-drinkers" it compared against were contaminated with people who had *stopped* because they were already ill.
- **But the absolute risk of one drink a week is small.** There is also no *safe* dose of cycling, or sunlight, or getting into a car. So the honest formulation keeps the truth and drops the overclaim: *there is no dose at which alcohol is good for you — there never was — and the risk starts at your first drink and rises with every one after it.*

Alcohol: no safe dose (cont.)

How today bends future risk: because the line has no threshold, risk accumulates with total lifetime intake rather than waiting for some "safe limit" to be crossed. The lever is the long-run pattern, not any single night — and the strongest honest illustration is that breast-cancer risk rises measurably per daily unit, with no floor (see the evidence).

Gambling: designed so you lose

The arithmetic needs no p-value and is unanswerable: the negative expected value of a gambling product is a **published design parameter**, not an empirical finding. The operator chose it in advance and printed it in the terms. Play long enough and variance washes out; the edge does not.

Gambling: designed so you lose (cont.)

On top of that sits a second, weaker, epidemiological claim — that the resulting harm is large and tracks the product, not the person — and it is graded honestly in the evidence panel: harm is measurably higher for the fastest, most continuous products (online slots) than for slower ones (sports betting), the same way tar delivery tracks the cigarette, not the smoker.

Gambling: designed so you lose (cont.)

How today bends future risk: gambling spend shows a dose-responsive association with financial distress and higher subsequent mortality, and adolescents are already in the exposure data. The single most protective move is upstream — recognising that the machine's edge is fixed and published, and not opening the app in the first place. The causal claims about debt, relationship breakdown and suicide are real but confounded; the arithmetic is not confounded by anything.

Eat food. Mostly plants. Not too much.

Start with the part that is strong, because it is unusually strong for nutrition. **Excess adiposity causally raises the risk of type 2 diabetes and cardiovascular disease** — Mendelian-randomisation studies, using genes fixed at birth, show this is causal rather than confounded. The IARC working group judged the evidence sufficient that excess body fatness **causes at least a dozen cancers**. And all-cause mortality rises with BMI above the mid-20s. So "why it matters" is not in doubt.

Eat food. Mostly plants. Not too much. (cont.)

The other strong, and under-appreciated, fact is about **reversibility**: the body defends a fat mass once it is reached. After substantial weight loss, resting metabolic rate stays suppressed and the weight is usually regained; and obesity tracks powerfully from childhood — a child with obesity is roughly five times as likely to carry it into adulthood, and the great majority of adolescents with obesity stay that way. The practical upshot is stark: **the biggest risk factor for carrying excess weight is having carried it already.** Prevention beats reversal by a wide margin, which is why the leverage is early and environmental.

Eat food. Mostly plants. Not too much. (cont.)

Two environmental levers on *how much* you eat have unusually clean experimental support: how **processed** the food is (people eat more of it, unprompted) and how **large the portion** is (serve more and people eat more, with no matching rise in fullness). Both are about changing defaults, not out-arguing appetite — a smaller plate and a less-processed cupboard do work that willpower rarely wins.

Eat food. Mostly plants. Not too much. (cont.)

Where the evidence is *weak* is the next question — *which named diet* to prescribe. Michael Pollan's *eat food, mostly plants, not too much* works less as a proven prescription than as a rule about **what to buy**. So the honest position is: the stakes are high and well-evidenced, and the defaults governing how much you eat are worth changing — but no specific named diet is a proven fix, and using smaller portions as a deliberate weight-loss tool meets the same defended-mass wall as everything else, so the effect on intake is far better evidenced than any lasting effect on the scale.

Move more, sit less

Narrowly: physical activity lowers mortality; the benefit is largest for people moving *out of the lowest* activity band; and the dose that captures most of it is far below the round numbers in circulation.

The dose-response is steeply inverse and **front-loaded** — the largest reduction in mortality comes from leaving the bottom quartile, measured with accelerometers so it isn't inflated by self-report. Getting off the floor matters far more than optimising the top.

Move more, sit less (cont.)

How today bends future risk: the honest causal signal (reverse causation is the obvious worry — ill people move less) is supported by Mendelian-randomisation work using accelerometer-derived phenotypes, which points to a genuine protective effect. Do not oversell the top of the curve, where confounding is heavy; the claim is strongest exactly where it matters most for a population — at the bottom, where the effect is largest and the causal support best.

Move more, sit less (cont.)

And the benefit is not only on the seventy-year clock. Physical activity is one of the best-evidenced near-term levers on mood, reliably easing symptoms of depression and anxiety over weeks — and a Mendelian-randomisation study supports a *causal* effect of activity on lower depression risk, since genotype is fixed at birth and cannot be explained away by "happier people move more." That near-term, causal wellbeing effect is why this page sits on both clocks.

Sleep like it matters

Narrowly: adequate sleep reliably improves mood, attention and emotional regulation over days-to-weeks — that part is experimental — and adolescent late sleep timing is biology, not laziness. The long-term "sleep saves your life" claim is much weaker, and we do not lean on it.

Sleep like it matters (cont.)

The honest case for sleep is the **two weeks, not the fifty years**. Short-term experimental sleep restriction reliably degrades attention, mood and emotional regulation; that is the immediately relevant, well-evidenced part. The environmental lever is real too: a delayed school start produced objectively more sleep, measured by actigraphy, with better grades and attendance.

Sleep like it matters (cont.)

How today bends future risk: honestly, the long-term causal story is weak — the U-shaped associations between sleep duration and mortality are badly confounded by reverse causation, because illness both shortens and lengthens sleep, and there will never be a randomised trial of lifetime sleep. So the durable move is to treat sleep as a daily performance-and-mood lever and protect its conditions (light, timing, a phone that isn't in the bed), rather than to frame it as a distant survival threat.

Slip, slop, slap

Narrowly: regular sunscreen use reduces skin cancer, and — unusually for a lifestyle exposure — this rests on a genuine randomised controlled trial rather than observation. The line itself is a model of the form: three verbs, an action for each, no threat and no blame.

The Nambour trial randomised a whole population to daily versus discretionary sunscreen, and ten years *after the intervention stopped* the daily-sunscreen arm had roughly half the melanomas. That persistence — benefit outlasting the behaviour — is the striking part, and it is why this is the strongest design on the site.

Slip, slop, slap (cont.)

How today bends future risk: melanoma is already among the commonest cancers in your age band, so this is one of the very few items where the ten-year and seventy-year clocks agree; and because damage accumulates, the protective habit pays both now and later. Be honest about the limits, though — it is one trial, one fair-skinned high-UV population, melanoma was a secondary endpoint, and the invasive-melanoma result rests on very small numbers. Even the best study here does not deliver certainty.

Change the rules, not just yourself

This page is the logical endpoint of everything else here — and it points at your ballot. If how much people smoke, drink, eat and move is powerfully shaped by their environment, then the highest-leverage health action is often to **change the environment**, and at population scale that means policy, decided collectively.

Change the rules, not just yourself (cont.)

The evidence is strongest exactly where the measure is structural and the study design is clean. Scotland's indoor smoking ban was followed by a sharp, replicated fall in heart-attack admissions, most of it in non-smokers who had merely been breathing the air. Its minimum unit price for alcohol was followed by a marked reduction in alcohol-attributable deaths, concentrated in the most deprived communities. The UK's soft-drinks levy triggered large-scale reformulation — the sugar came out of the drinks — with a targeted reduction in childhood obesity in the poorest areas. (The precise figures, with sources, are in the evidence panel below.) Be honest about the weaker case, too: **active-travel infrastructure** — cycle paths, safe routes — is suggestive, not proven.

Change the rules, not just yourself (cont.)

Put those together and an honest conclusion follows: the biggest population health improvements have generally come from changing the shared environment, not from exhorting individuals — so **engaging as a citizen, and voting for representatives who prioritise health-promoting environments over ill-health-promoting ones, is itself a high-leverage, evidence-based health act.** This is a reference, not a campaign, so it stops there: the evidence for *what works* is above; *which* measures are worth their trade-offs, and *who* best represents them, is a question of values that is yours to answer, not ours.

Trust, but verify.

Full evidence — with the study designs, the effect sizes, and the honest weaknesses — is on the site.

Information is power.